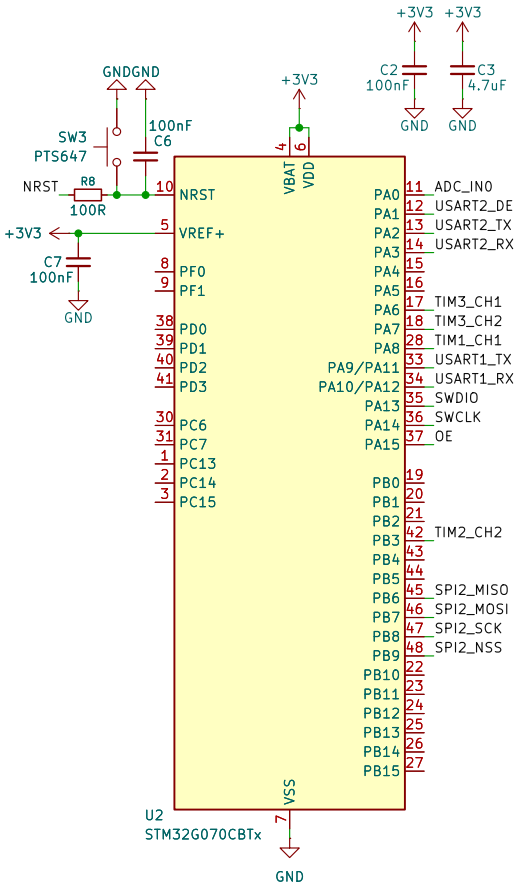
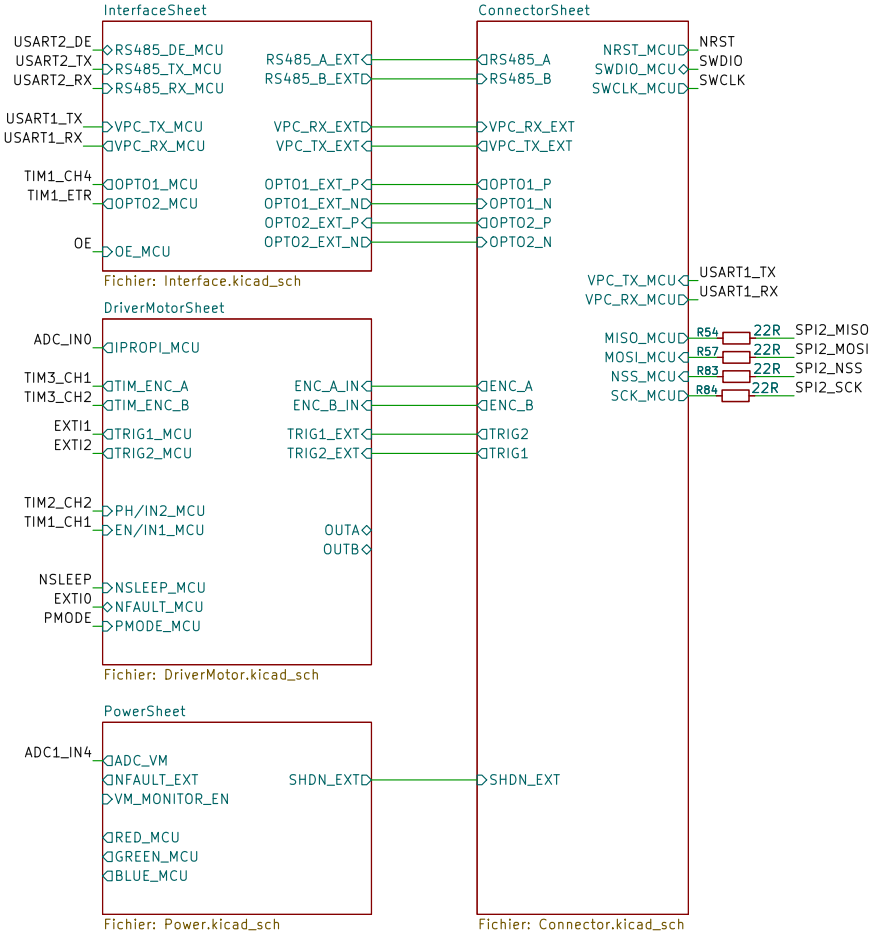
Microcontroller Core & Supervision

32-bit ARM Cortex-M0+ MCU (STM32G070KBT6) for motor control, sensor acquisition, and RS-485 telemetry.



- Clock: Internal RC up to 64 MHz | Supply: +3.3V
- ADC (PA0): Motor current sensing (IPROPI, 0V to 3.3V)
- Reset: Active-low with hardware RC filter (100R / 100nF, τ = 10 µs)



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Sheet: /
File: KinetiCoreDC.kicad_sch

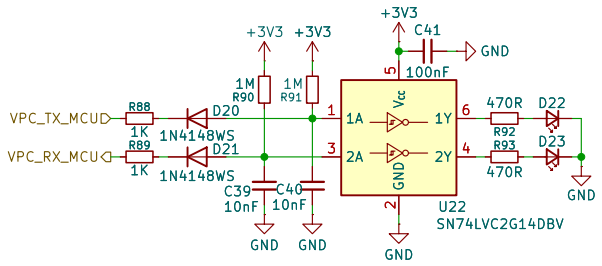
Title: KinetiCoreDC – Main

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.6			Id: 1/5

Text

UART activity LED indicators (pulse stretcher)

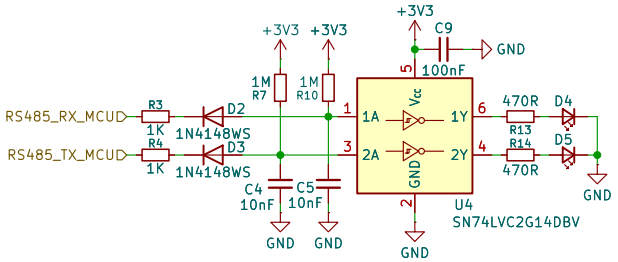
Asymmetrical RC Monostable (10 ms Hold Time) & Schmitt-Trigger Buffer



- Monostable hold time: τ ≈ 10 ms (1MΩ / 10nF + fast discharge diode)
- High-impedance tap: Does not load UART RX/TX lines

UART activity LED indicators (pulse stretcher)

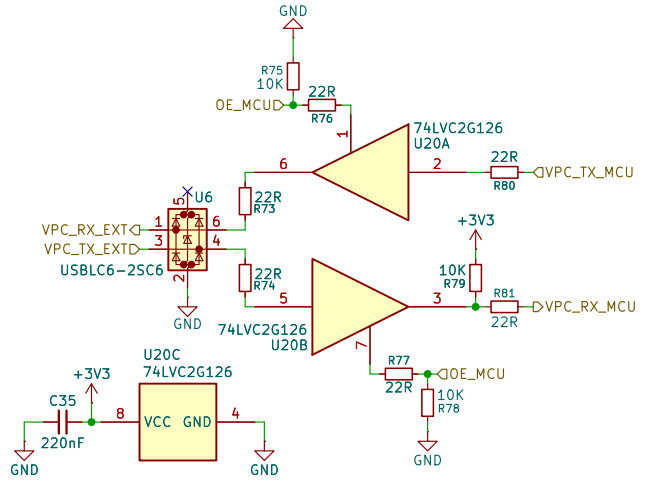
Asymmetrical RC Monostable (10 ms Hold Time) & Schmitt-Trigger Buffer



- Monostable hold time: τ ≈ 10 ms (1MΩ / 10nF + fast discharge diode)
- High-impedance tap: Does not load UART RX/TX lines

Buffer LVC Interface (Protected UART)

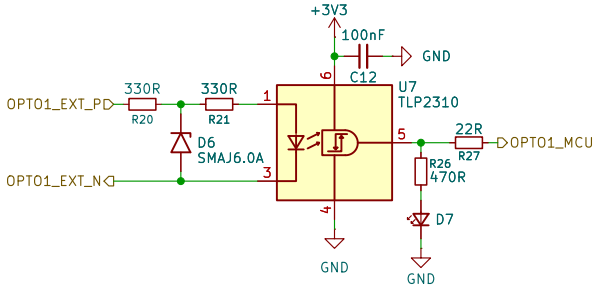
3-State Isolation (74LVC2G126), ESD Clamping & Current Limiting



- Lines that can be disabled (Hi-Z) via OE_MCU (active high, with built-in 10k pull-downs).
- 10k pull-up on RX_MCU to ensure the idle level (3.3V) when the buffer is disabled.

Optoisolated PWM Input N°1

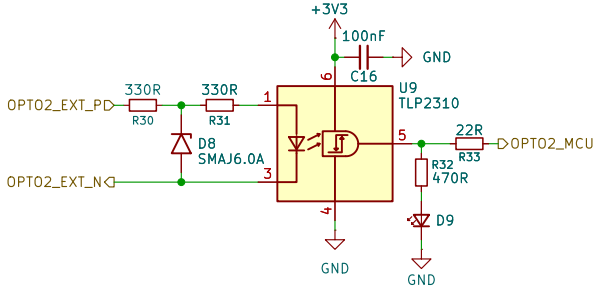
Optoisolated input for secure PWM/GPIO control, protecting the logic circuit from DC motor noise.



- Input range: 3V3 to 5V differential (R_series = 660Ω, ~2.73mA @ 3V3)
- Max frequency: 5 Mbps (Push-pull output, active-high)
- Protection: Reverse/Overvoltage clamp at 6.0V (SMAJ6.0A)

Optoisolated PWM Input N°2

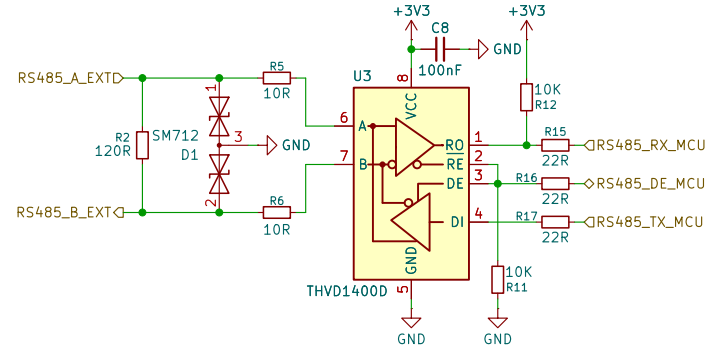
Optoisolated input for secure PWM/GPIO control, protecting the logic circuit from DC motor noise.



- Input range: 3V3 to 5V differential (R_series = 660Ω, ~2.73mA @ 3V3)
- Max frequency: 5 Mbps (Push-pull output, active-high)
- Protection: Reverse/Overvoltage clamp at 6.0V (SMAJ6.0A)

Protected Half-Duplex RS-485

Protected 3.3V half-duplex transceiver stage (up to 500 kbps)



- Bus specs: Half-duplex up to 500 kbps (Slew-rate limited, 120Ω terminated)
- Common-mode protection: −7V to +12V (SM712 TVS + 10Ω series)
- Control: DE/RE tied together (Default RX mode via 10kΩ pull-down)

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Sheet: /InterfaceSheet/
File: Interface.kicad_sch

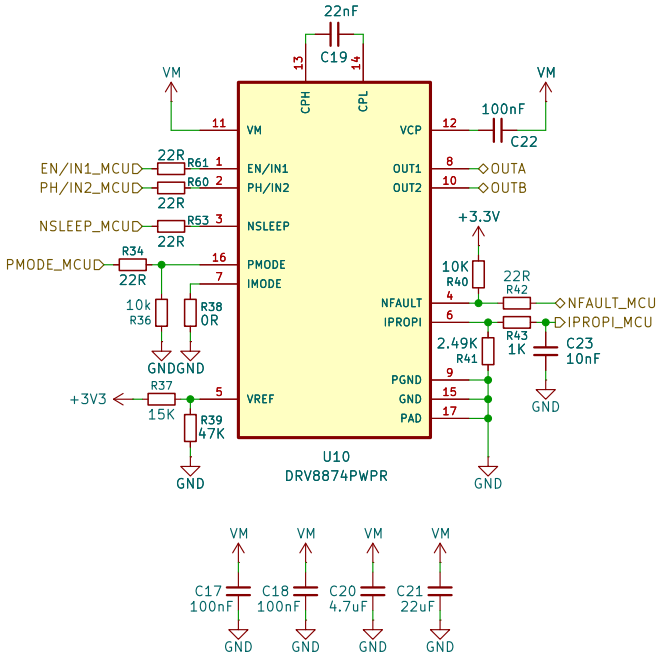
Title: KinetiCoreDC – Interface

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.6			Id: 2/5

Text

H-Bridge Motor Driver

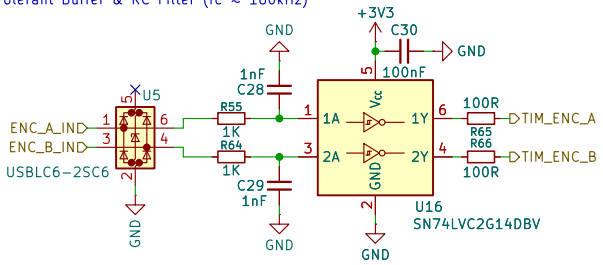
Integrated DC motor driver with internal current sensing and regulation



- Operating range: VM = 5.1V to 18V | I_peak = 6A
- Current limit (I_TRIP): ~2.6A (VREF = 2.50V via 15k/47k divider, R_IPROPI = 2.49kΩ)
- Current sense (IPROPI): 455 µA/A → ~1.13 V/A (RC filter fc ≈ 16 kHz)
- Control mode: PMODE selectable | NFAULT: Open-drain (10kΩ pull-up)

Encoder interface & ESD protection

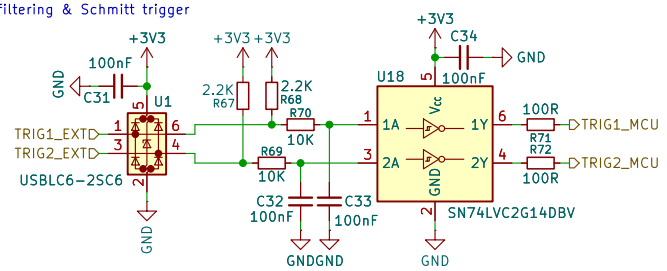
Incremental Quadrature (A/B),
5V/3V3 Tolerant Buffer & RC Filter (fc ≈ 160kHz)



- Input: Quadrature A/B, 3.3V / 5V tolerant
- HF Noise Filter: Low-pass RC (1kΩ / 1nF, fc ≈ 160 kHz, τ = 1 µs)
- Output: Inverted Schmitt-trigger to STM32 TIM3_CH1/CH2

Limit Switch Input Conditioning

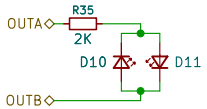
ESD protected limit switch interface
with RC filtering & Schmitt trigger



- Input type: Dry contact to GND (Internal 2.2kΩ pull-up to +3.3V)
- Debounce filter: Low-pass RC (10kΩ / 100nF, fc ≈ 160 Hz, τ = 1 ms)
- Output: Inverted Schmitt-trigger logic (Active-HIGH at MCU when switch closed)

Output Status Indicator

Bi-directional LED polarity and activity indicator
across H-bridge outputs



- Antiparallel LEDs across OUTA/OUTB (2kΩ ballast)
- Indicates rotation direction (polarity) and PWM duty cycle (brightness)

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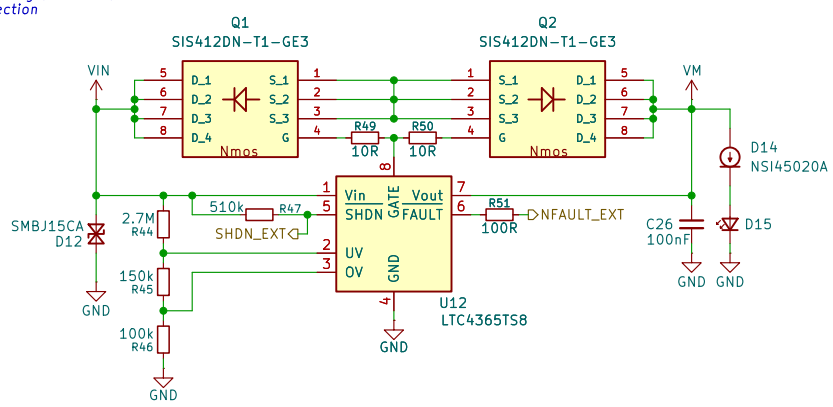
Sheet: /DriverMotorSheet/
File: DriverMotor.kicad_sch

Title: KinetiCoreDC – Driver Motor

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.6			Id: 3/5

External Supply Protection

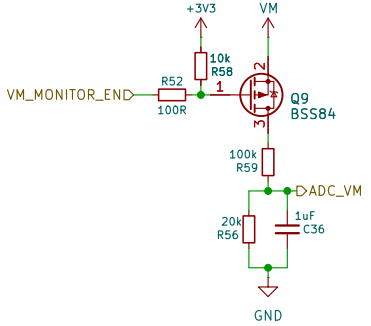
Overvoltage, Undervoltage, Reverse, and ESD Input Protection



- Window thresholds (LTC4365, Recovery delay: 36 ms, 5% hysteresis, Max 16V for MCP1703 LDO):
  - [Default] 2S & 3S Compatible: UVLO = 5.90V (6.20V rise) | OVLO = 14.75V (14.01V fall) -> R_top=2.7MΩ, R_mid=150kΩ, R_bot=100kΩ
  - 2S LiPo Optimized (7.4V): UVLO = 6.45V (6.78V rise) | OVLO = 10.84V (10.30V fall) -> R_top=2.0MΩ, R_mid=68kΩ, R_bot=100kΩ
  - 3S LiPo Optimized (11.1V): UVLO = 9.15V (9.61V rise) | OVLO = 14.28V (13.57V fall) -> R_top=2.7MΩ, R_mid=56kΩ, R_bot=100kΩ
  - 9V Battery (6LR61 / Li-Ion): UVLO = 5.50V (5.78V rise) | OVLO = 11.00V (10.45V fall) -> R_top=2.0MΩ, R_mid=100kΩ, R_bot=100kΩ
- Protection: Reverse polarity to −40V (Back-to-back N-MOS) & 30V TVS clamp
- SHDN_EXT: Active-LOW (<0.4V = Sleep <10µA, 510kΩ pull-up to VIN)
- NFAULT_EXT: Open-drain active-LOW (Requires external pull-up on host board)

Battery Monitor

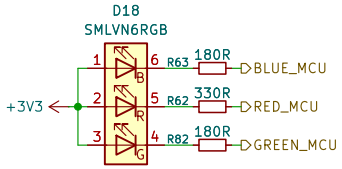
Switched Zero-Leakage Battery Level



- Power Gating: BSS84 (Q9) P-MOSFET with 10kΩ pull-up (R58) to +3V3; eliminates divider quiescent leakage via VM_MONITOR_EN.
- Divider Network: 100kΩ (R59) / 20kΩ (R56) yielding a division ratio of 1/6 (V_ADC = 0.167 × VM).
- Filter & Buffer: 1µF (C36) low-pass / charge-reservoir capacitor at ADC_VM pin (τ ≈ 16.7 ms, fc ≈ 9.5 Hz).
- Operating Range: Scaled for battery inputs up to 19.8V (V_ADC ≤ 3.3V), compatible with multi-cell Li-ion tracking.

RGB Led Indicator

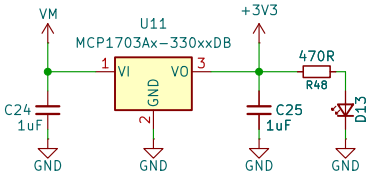
Status LED – Active-LOW Sink Control



- Common Anode (+3.3V) | Active-LOW sink to MCU PWM pins
- Red: VF ≈ 2.0V | R = 330 Ω -> IF ≈ 3.9 mA
- Green: VF ≈ 2.85V | R = 180 Ω -> IF ≈ 2.5 mA
- Blue: VF ≈ 2.85V | R = 180 Ω -> IF ≈ 2.5 mA
- Balanced current ratios for natural white point (< 9 mA total)

Supply LDO Regulator

Low quiescent current 3.3V linear regulator for digital control logic



- Output: +3.3V ±0.4% | I_max = 250 mA | I_q = 2 µA
- Input: Supplied from protected VM rail (Max 16V continuous)

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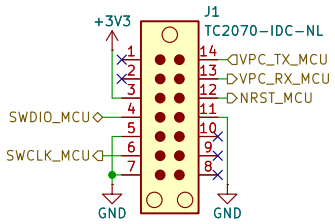
Sheet: /PowerSheet/
File: Power.kicad_sch

Title: KinetiCoreDC – Power

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.6			Id: 4/5

SWD Programming & UART Debug Interface

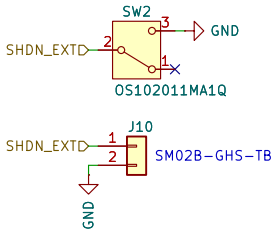
SWD flashing and VCP serial console connector
compatible with STDC14 / STLINK-V3



- Footprint: Tag-Connect TC2070-IDC-NL (No-Legs, spring-pin probe)
- Interfaces: SWD (SWDIO/SWCLK) + VCP UART (USART2) + Target NRST

Manual Power Enable

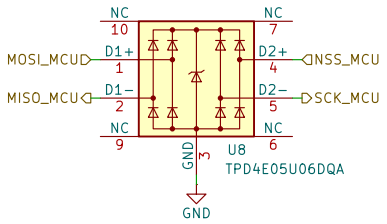
On-board SPDT slide switch for manual power shutdown and sleep mode (< 10 µA).



- ON (Pin 1 – NC): Pulled HIGH to VIN via 510 kΩ (Normal operation)
- OFF (Pin 3 – GND): Pulls SHDN_EXT LOW (< 0.4 V), disabling input MOSFETs (< 10 µA sleep)

SPI/I²C ESD protection

4-Channel Ultra-Low Capacitance Transient Voltage Suppressor



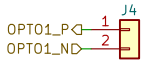
- Technology: 4-channel bidirectional ESD protection array (USON-10 flow-through package)
- Electrical Specs: V_RWM = 5.5 V | V_BR ≥ 6.5 V | I_off = 100 nA max (no VCC/VBUS pin required)
- Parasitic Capacitance: C_IO ≈ 0.5 pF (ultra-low load preserving fast SPI SCK & I²C rise times)
- ESD Level: ±12 kV Contact Discharge, ±15 kV Air-Gap (IEC 61000-4-2 Level 4)

I/O & Field interface connectors

Ultra-Compact Wire-to-Board Interface (JST-GH Series / 1.25mm Pitch)

Opto Connector n°1

SM02B-GHS-TB



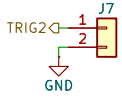
Opto Connector n°2

SM02B-GHS-TB



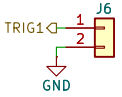
Limit Switch n°2

SM02B-GHS-TB



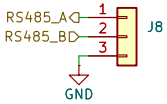
Limit Switch n°1

SM02B-GHS-TB



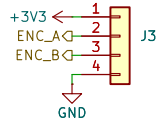
Half-Duplex RS-485

SM03B-GHS-TB



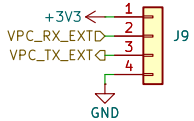
Encoder

SM04B-GHS-TB



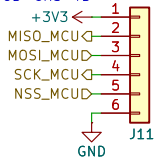
UART

SM04B-GHS-TB



SPI/I²C

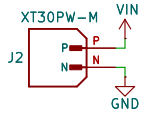
SM06B-GHS-TB



- Electrical Specs: 1.0 A max per pin | 50 V AC/DC | AWG 26-30
- Mechanical Retention: Active positive lock (latch click) with high vibration resistance
- Form Factor: Low-profile 1.25 mm pitch SMT | Dual solder reinforcement tabs
- Application Scope: Differential RS-485, UART debug, encoder inputs & limit switches (not for motor power)

Supply Connector

High-Current DC Input (XT30PW-M)



- Specs: 15 A cont. (30 A peak) | Gold-plated (< 0.8 mΩ)
- Keying & Fit: Polarized keyed | Low-profile right-angle mount
- Interface: Anti-vibration friction lock | 2S-4S LiPo standard

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Sheet: /ConnectorSheet/
File: Connector.kicad_sch

Title: KinetiCoreDC – Connector

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.6			Id: 5/5

Text